Octave magnet

Does a bar magnet have four pairs of poles, as Walter Russell's painting says? The exact field of a bar, the bench test that decides, and the pattern closed into a ring of twelve sections (register entry H33)

kronspice simulation report · 2026-11-07 · recorded run: run_output.txt (a few seconds)

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Summary

Hypothesis H33 (user hypothesis, 2026-11-05) comes from Walter Russell's painting "Analysis of the Octave Bar Magnet", which says a bar magnet has *"four pairs of poles — not one pair"*, drawn as a chain of lens-shaped rings strung along the magnet's axis. The example builds the standard field of a bar magnet exactly, shows what each claim looks like as a field, and fixes the bench measurement that decides between them. A second part bends the pattern into a ring magnet of twelve 30° sections. This PDF covers the example's two reports: the bar (Parts 1–3) and the ring (Part 5).

- **The model is exact to round-off:** the current-sheet field equals the solenoid formula on the axis to 1.3×10^{-6} of $\mu_0 M_0$; the loop field equals Biot–Savart off the axis to 2.160×10^{-6}; a kronspice search coil reads the same flux as $\Sigma M\cdot I$ and $2\pi\psi$ to six digits.
- **One pair against four:** a uniform bar has 2 magnetic-charge sites and gives 2 sign changes of B_z 3 mm beside it; four reversed segments (S N | N S | S N | N S) have 5 sites and give 5.
- **The lenses are real field lines** of a chain of poles: the four-segment bar draws the chain of lenses in Russell's painting; the uniform bar's outside lines run end to end in one family.
- **The decisive test is a Hall-probe scan** of an ordinary bar: two sign changes support the standard prediction, five or more on an unmodified bar support Russell. The model cannot decide which a real bar is.
- **The ring of twelve works when built that way:** both alternating rings (radial, or along the ring with like poles meeting at the joints) show 12 sign changes round the outside (peaks 291 and 315 mT); the all-outward control shows 0.
- **Status: untested** for the bar, until the measurement is made.

What is being tested

Hypothesis (H33). A bar magnet has four pairs of poles, not one; its field is a chain of lens-shaped rings on the axis. **Standard prediction.** A uniformly magnetized bar has one pair of poles (magnetic charge on the end faces only) and its outside field lines run end to end in one family. Four pairs, as a field, need three reversals of M inside the bar (five charge sites end to end, eight if four magnets are separated) and give at least five sign changes of B_z beside it.

The painting itself is described in doc/russell-octave-report.md; it is not reproduced here. That note also records where the painting disagrees with measurement: Russell's structures repeat in octaves (frequency ratio 2), while hydrogen's lines follow $1/n^2$ (Lyman α to the Lyman limit is 4/3, Hα to Lyman α is 5.4).

Report 1 — The bar magnet (Parts 1–3)

The model

- **The bar as currents.** A magnetization $M(z)\hat{z}$ in a rod of radius a is exactly a sheet of azimuthal current $M(z)$ on its side ($J = \mathrm{curl}\, M$, $K = M\times n$, n the outward unit normal; Jackson §5.8–5.9), built as 2,000 coaxial loops each carrying $M\cdot dz$.
- **Poles** are the magnetic charge $-\mathrm{div}\, M$: the end faces, plus every place inside where M changes.
- **Field lines** are the contours of the flux function $\psi = \rho A_\varphi$, summed from the exact elliptic-integral potential of a loop (Smythe §7.10).

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- **The bar:** an alnico rod, 12.7 mm × 76.2 mm, B_r 1.25 T ($M_0 = B_r/\mu_0 = 9.947\times10^5$ A/m). Demagnetization is ignored, declared.

Checks

Check	Result
Sheet field on the axis (2000 loops) against the exact solenoid formula	worst 1.3×10^{-6} of $\mu_0 M_0$
One loop off the axis, elliptic formula against Biot–Savart (em.Loop, 2000 segments)	B_ρ 2.160×10^{-6}, B_z 3.097×10^{-7}
kronspice search coil: flux $\int V\,dt$ against $\Sigma M\cdot I$ and $2\pi\psi$	equal to six digits

Poles as magnetic charge

Bar	Charge sites (position, charge)
uniform (one pair)	2: −38.1 mm (−126 A·m), +38.1 mm (+126 A·m)
four reversed segments (four pairs)	5: −38.1 mm (−126 A·m), −19.1 mm (+252 A·m), +0.0 mm (−252 A·m), +19.1 mm (+252 A·m), +38.1 mm (−126 A·m)

The uniform bar carries charge only on its two end faces: one pair. Four pairs need M to reverse three times inside; then S N | N S | S N | N S gives the two ends plus three doubled "consequent" poles, four dipoles end to end.

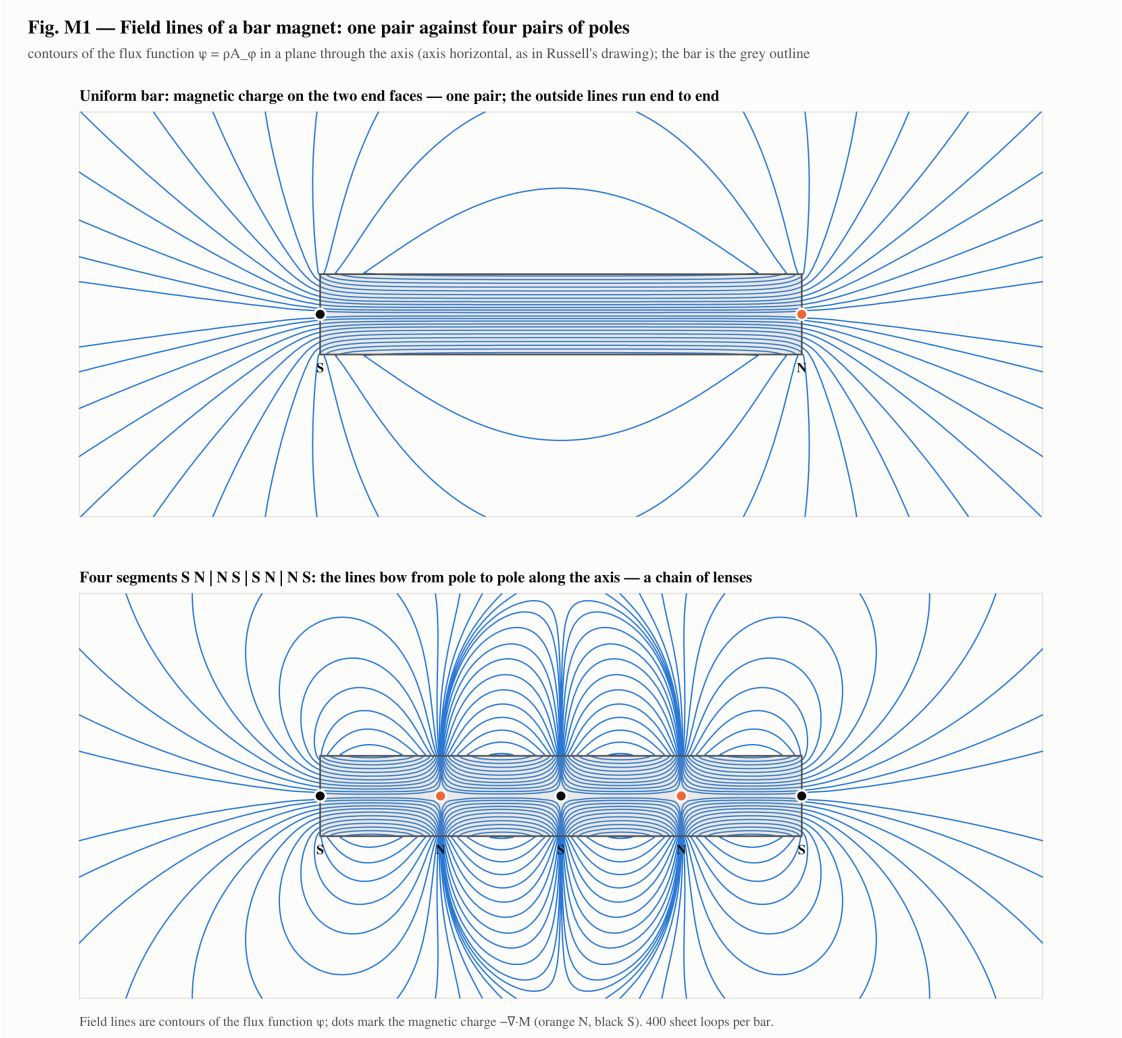


Figure 1 (M1). Field lines (contours of $\psi = \rho A_\varphi$) in a plane through the axis (axis horizontal, as in Russell's drawing); the bar is the grey outline and dots mark the magnetic charge (orange N, black S). Top: uniform bar, charge on the two end faces, outside lines end to end. Bottom: four segments S N | N S | S N | N S, the lines bow from pole to pole along the axis, a chain of lenses.

A field line that leaves the axis at one pole and returns at the next is a lens strung on the axis, so a chain of poles draws a chain of lenses, as in Russell's painting.

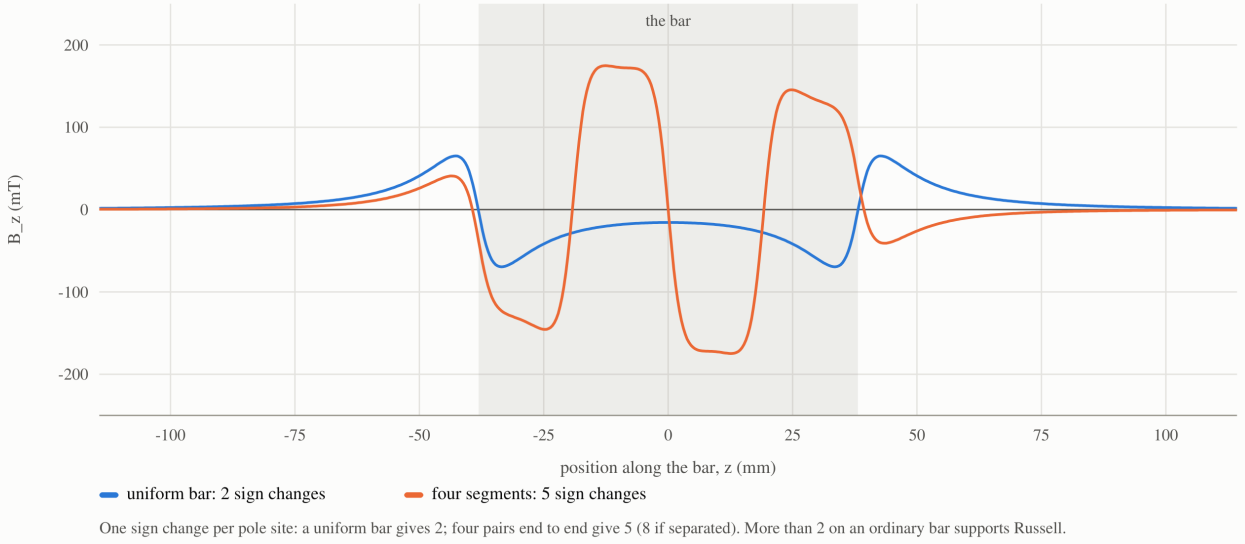
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The bench test

Bar	Charge sites	B_z sign changes on the axis	B_z sign changes 3 mm beside it
uniform (an ordinary bar): one pair	2	0	2
four segments S N | N S | S N | N S: four pairs end to end	5	3	5

Fig. M2 — The bench test: B_z along a line 3 mm beside the bar

a Hall probe run parallel to the axis; sign changes count the poles it passes



One sign change per pole site: a uniform bar gives 2; four pairs end to end give 5 (8 if separated). More than 2 on an ordinary bar supports Russell.

Figure 2 (M2). B_z along a line 3 mm beside the bar (a Hall probe run parallel to the axis); the shaded band is the bar. Uniform bar: 2 sign changes. Four segments: 5. One sign change per pole site.

Search coil in kronspice (a ring round the bar, radius a + 3 mm), flux from ∫V dt against ΣM·I and 2πψ:

z	circuit ∫V dt	mutual inductances ΣM·I	2πψ
0.0 mm	1.53867×10^{-4} Wb	1.53867×10^{-4} Wb	1.53867×10^{-4} Wb
19.1 mm	1.49659×10^{-4} Wb	1.49659×10^{-4} Wb	1.49659×10^{-4} Wb
38.1 mm	7.85866×10^{-5} Wb	7.85866×10^{-5} Wb	7.85866×10^{-5} Wb
57.2 mm	7.28009×10^{-6} Wb	7.28009×10^{-6} Wb	7.28009×10^{-6} Wb

Bench protocol for the bar (from report.html and doc/russell-octave-report.md):

- Take an ordinary bar magnet and declare its measured B_r.
- Run a Hall probe (or the search coil modelled here) parallel to the axis at a fixed gap, from beyond one end to beyond the other, in about 1 mm steps.
- Count the sign changes of B_z. Two supports the standard model; more than two on an unmodified bar supports Russell (five for four pairs end to end, eight if separated).
- Compare the trace point by point with the example's scan for the declared B_r.

Report 2 — The ring of twelve sections (Part 5)

The octave bar bent into a ring: a ring magnet cut into twelve 30° sections, each magnetized opposite to its neighbours (user request, 2026-11-05). Two ways of making neighbours opposite are built, with a control:

- **Radial.** Each section is magnetized along its own centre line, outward and inward in turn, so the outer face reads N S N S … — the standard multipole ring of motors, magnetic couplings and encoders.
- **Along the ring.** Each section is magnetized along the ring, clockwise and anticlockwise in turn, so like poles meet at every joint (N|N, S|S): the four-segment bar closed into a ring, with consequent poles at the joints.
- **Control.** All twelve sections magnetized outward.

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The ring is 50 mm across, 30 mm bore, 10 mm tall, B_r 1.25 T (declared); each section is uniformly ("parallel") magnetized, as segmented magnets are made. A uniformly magnetized body carries all its magnetic charge on its surface, $\sigma = M \cdot n$ (n the outward unit normal; Jackson §5.9): on the outer arc, the inner arc and the two cut faces; the flat top and bottom carry none. Outside, $B = \mu_0 H$ is the Coulomb field of that charge. Twin: the same volume as point dipoles $m = M \cdot dV$ (Jackson eq. 5.56). Field lines are traced by RK4 along the unit vector of B from the outer N faces, at equal steps of flux, until they reach the magnet again.

Check	Radial	Along the ring
Surface charge against the point-dipole twin, 6 points outside	7.8×10^{-5} of |B|	7.9×10^{-5}
Net magnetic charge, of Σ|Q|	4.8×10^{-17}	2.2×10^{-17}
div B and curl B outside (central differences, of the field gradient)	1×10^{-7}, 3×10^{-7}	3×10^{-7}, 1×10^{-7}

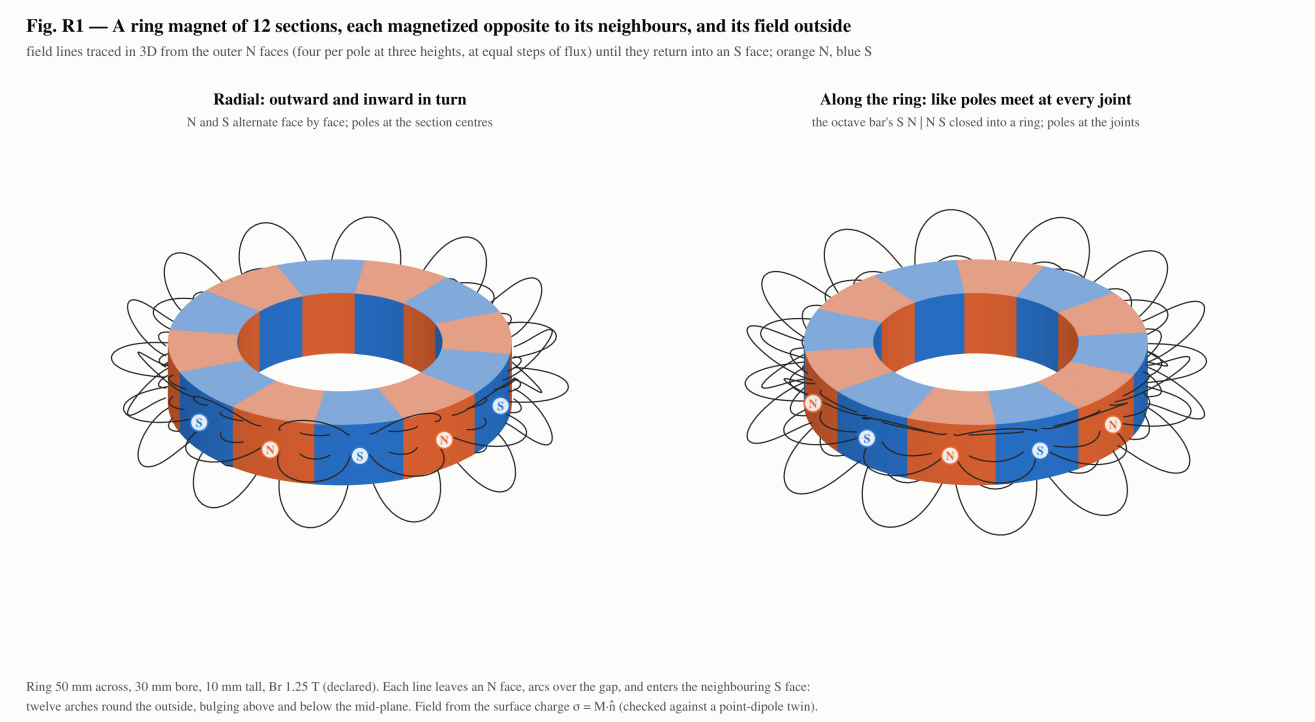


Figure 3 (R1). The ring in 3-D, orange N and blue S, with field lines traced from the outer N faces (four per pole at three heights) until they return into an S face. Left: radial, poles at the section centres. Right: along the ring, like poles meeting at every joint. Each line arches over to the neighbouring S pole: a crown of twelve arches, flat at mid-height and bulging above and below it.

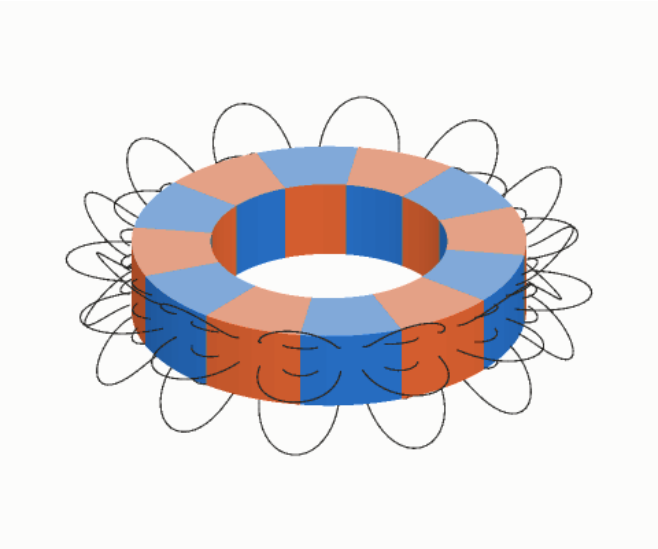


Figure 4. First frame of the turntable animation viz/ring3d_radial.gif: the radial ring, poles at the middle of each section.

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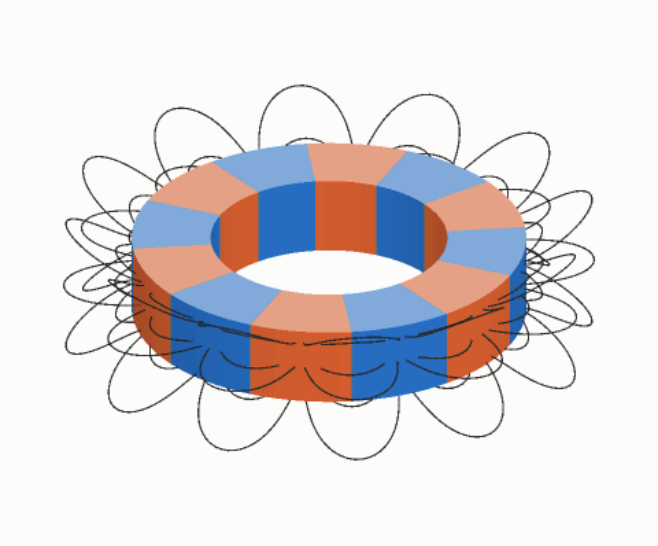


Figure 5. First frame of viz/ring3d_tangential.gif: the ring magnetized along the ring; each section is N at one end and S at the other, so the poles sit at the joints.

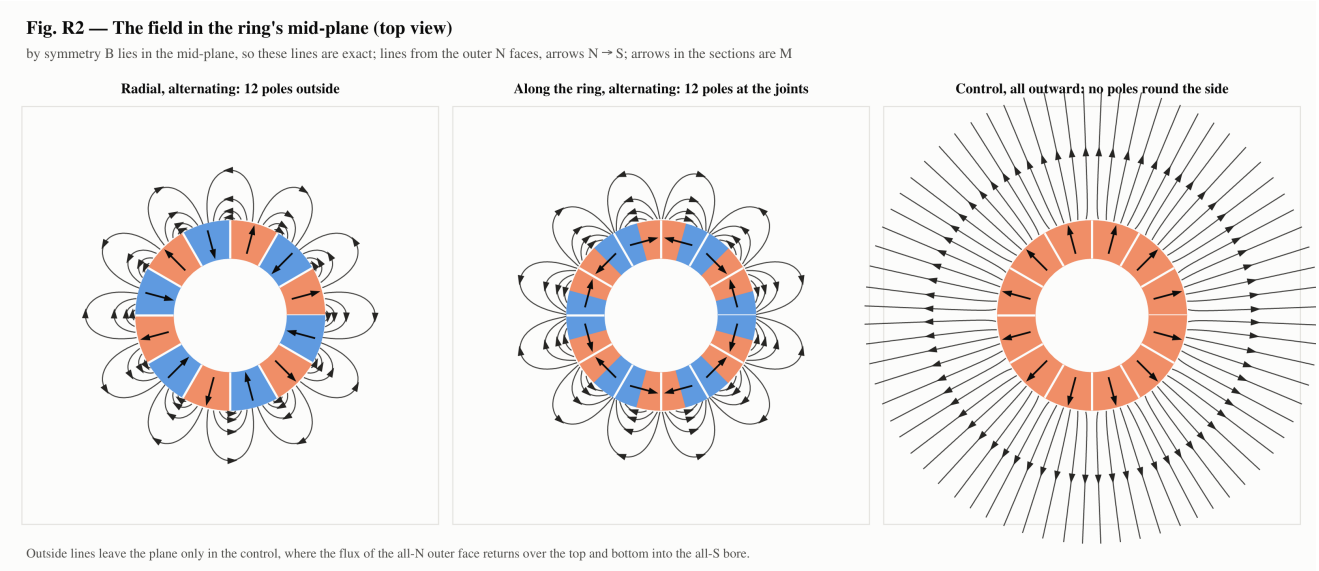


Figure 6 (R2). The field in the mid-plane, seen from above; by symmetry B lies in that plane, so these lines are exact. Arrows in the sections show M. Left: radial, 12 poles outside. Middle: along the ring, 12 poles at the joints. Right: control, all outward; its outer face is all N and its bore all S, so its flux leaves radially and returns over the top and bottom into the bore, and no lines close in the plane.

The field round the outside

B_r 3 mm outside the outer face, in the mid-plane: the scan a Hall probe makes when carried round the ring.

Ring	Sign changes	Peak $\lvert B_r \rvert$	N poles at
radial, alternating (N S N S … outside)	12	291 mT	15°, 75°, 135°, 195°, 255°, 315° (section centres)
along the ring, alternating (N|N, S|S joints)	12	315 mT	30°, 90°, 150°, 210°, 270°, 330° (joints)
radial, all outward (control)	0	223 mT	none (B_r > 0 all round)

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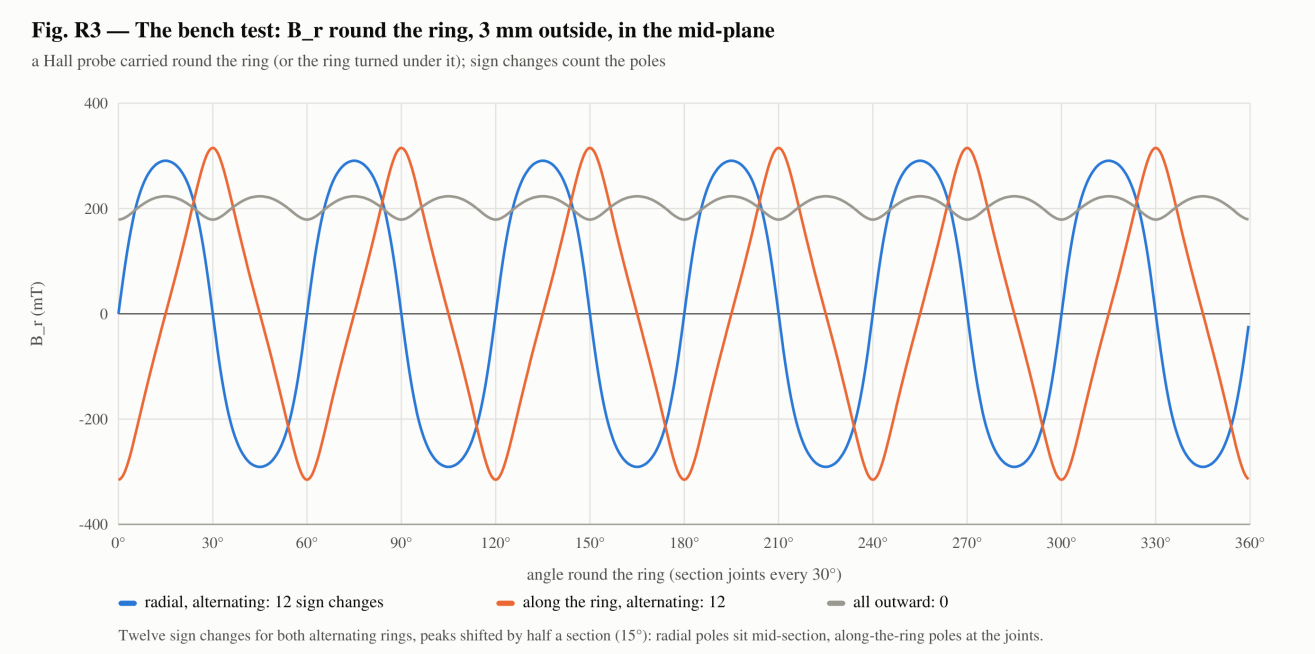


Figure 7 (R3). B_r round the ring, 3 mm outside, in the mid-plane. Twelve sign changes for both alternating rings, their peaks shifted by half a section (15°); none for the all-outward control.

Both alternating rings have twelve poles round the outside, six N and six S; the field falls off quickly with distance because neighbouring poles cancel. They look alike from outside; their poles sit 15° apart.

A pickup coil beside the turning ring, in kronspice

A 200-turn, 6 mm pickup coil faces the outer side, 1.5 mm out (R 8.26 Ω from the wire, L 246 µH from Wheeler's multilayer formula, both declared), driving an 8.2 Ω load while the radial ring turns at 3000 rpm.

Quantity	Result
Electrical cycles per revolution	6 (300 Hz), one per pole pair
Open-circuit EMF	6.23 V peak
Across the load	1.754 V rms, 375 mW
Ledger over two revolutions	in 3.01119×10^{-2} J (shaft, through the EMF) = heat 3.01119×10^{-2} J + Δ stored -1.5×10^{-7} J
Ledger residual	5.8×10^{-6}

Every joule the load receives is work done turning the ring; the coil's EMF source stands in for the shaft.

Validation

Test	Checks	Tolerance
TestSheetAxis	sheet field = exact solenoid formula on the axis, both bars	10^{-5} of $\mu_0 M_0$
TestLoopFieldTwin	elliptic loop field = Biot–Savart, four points	10^{-5}
TestPoles	2 charge sites (uniform), 5 (four segments), net charge zero	exact
TestScan	sign changes 2 / 0 (uniform), 5 / 3 (four segments)	exact
TestSearchCoil	kronspice fluxmeter = ΣM·I = 2πψ	10^{-6}
TestRingTwin	surface charge = point-dipole twin, 6 points outside, both rings	2×10^{-4}
TestRingChargeDivCurl	net charge zero; div B, curl B zero outside	10^{-12}; 10^{-5}
TestRingScan	12 / 12 / 0 sign changes; N poles at 15° + 60°k (radial), 30° + 60°k (along the ring)	exact; 0.6°
TestRingLinesClose	every line from an outer N face ends on an S section	exact
TestRingPickup	kronspice pickup coil, 6 cycles per revolution, ledger closes	exact; 10^{-4}

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Limits of the model

- Demagnetization is ignored and B_r is declared; a real bar's measured B_r must be declared for the bench comparison.
- The model computes either case exactly; it cannot say which one a real bar is. That is a measurement.
- The ring's sections are ideal uniform magnets; the pickup coil's R and L are declared, not measured.
- Lens shapes made from two centres, colour as state and the octave (ratio 2) in the painting have no counterpart in the model (doc/russell-octave-report.md).

Status

H33 is *untested*: the model and the bench protocol are built; the claim needs the Hall-probe scan of an ordinary bar. Russell's like-poles-meet pattern closed into a ring gives twelve real poles, as many as the standard radial multipole ring, but the pattern has to be built into the magnetization: a ring magnetized one way all round has no poles round its side. The bar's status is unchanged by the ring. The same words read as hydrogen are register entry H34 (*hydrogen_poles*).

Reproducing the results

```
go run ./examples/octave_magnet                # a few seconds
go run ./examples/octave_magnet -svg viz       # also the figures and the ring GIFs
go test ./examples/octave_magnet
python examples/octave_magnet/viz/make_report.py examples/octave_magnet/octave_magnet_report.pdf
```

Sources: W. Russell, "Analysis of the Octave Bar Magnet" (painting); *The Universal One* (1926); *The Secret of Light* (1947). J. D. Jackson, *Classical Electrodynamics*, 3rd ed., §5.5, §5.6 (eq. 5.56), §5.8–5.9; W. R. Smythe, *Static and Dynamic Electricity*, §7.10; H. A. Wheeler, *Proc. IRE* 16, 1398 (1928).

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